

BARD

BEHAVIOURALLY ADAPTIVE REACTIVE DIALOGUE
ERIMENTHA — AN LLM-DRIVEN RPG DEMO

Godot 4.6

GDScript

Python 3 Pipeline

Ollama · Gemma 4 E4B

Local GPU Inference

Academic Capstone 2025

OVERVIEW

BARD is an academic RPG demonstrating how a locally-running Large Language Model can generate unique, context-aware NPC dialogue each in-game day — creating the illusion that the world *remembers* what you did.

The player hunts monsters, completes bounties, and sleeps at the inn. Every night a Python pipeline queries a local LLM and rewrites each NPC's dialogue based on that day's events — before morning arrives.

3

NAMED NPCS

12

ENEMY TYPES

4

BOSS ENCOUNTERS

36

BOUNTY CONTRACTS

CORE GAMEPLAY LOOP

- Hunt** — Travel to Ashfield and clear monster-infested zones to fulfil bounty contracts
- Turn In** — Return to Thornwall, visit the Guild Commander to claim your reward in Scripts

THE LLM PIPELINE

The academic heart of BARD. Every night, a **Python subprocess** wakes, reads the day's game state, and prompts a locally-running LLM to write fresh dialogue for each named NPC — no cloud, no API keys.

```
Player sleeps at Inn
→ end_day() writes game_state.json
→ Godot spawns end_of_day.py subprocess
→ Pipeline reads kills, bounties, flags, NPC memory
→ Ollama (Gemma 4 E4B) generates dialogue JSON
→ Validates schema → repairs if malformed → fallback
→ Writes dialogue/{npc_id}_day{N}.json
→ LLM extracts one NPC recollection fact
→ Writes pipeline_ready.flag
→ Godot detects flag → reloads all dialogue → go to town
```

Communication model: Godot polls for sentinel flag files every 3 seconds — no blocking, no threads. A 400 px progress bar animates as each NPC is processed.

Ollama HTTP API

3× retry logic

JSON schema validation

LLM-based repair

Fallback chain

180 s timeout

BOUNTY SYSTEM

A contract loop that feeds the LLM, scoped to specific monster types and has rich, precise data to reference.

```
bounty_pool.json (36 static contracts)
→ Daily refresh: min(day, 3)
→ Player accepts → tracked in memory
→ Kills in zone increment contract
→ Turn in at Guild Commander
→ end_day() writes completed
```

Small contract	10 Scripts
Medium contract	25 Scripts
Large contract	50 Scripts

DIALOGUE ARCHITECTURE

LLM-generated content is surgical at runtime — preserving system reliability and narrative content.